10. Exponential Distribution.

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1. Key Concept

The exponential distribution models the time between events in a Poisson process, where events occur continuously and independently at a constant rate.

- **Parameter**: λ (rate parameter, same as in Poisson).
  - ↳ $\lambda = 0.2$ means 0.2 events per minute (or 1 event every 5 minutes on average)
- **Mean**: $1/\lambda$ (average time between events).
- **Variance**: $1/\lambda^2$

2. Probability Density Function (PDF)

→ PDF of the exponential distribution describes the likelihood of observing a specific time interval between events in a Poisson process.

Formula,

For $\lambda > 0$, the PDF is

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0.$$

where,

λ (Rate Parameter)
- ↳ Represent the average no. of events per unit time
- ↳ E.g. $\lambda = 0.2$ means 0.2 events per minute (or 1 event every 5 minutes on average)

$e^{-\lambda x}$ (Exponential Decay).
- ↳ ensures the probability decrease as the time interval 'x' increase.
- ↳ This matches the intuition that longer waiting times become less likely.

3. Example Calculation

For $\lambda = 0.2$ (mean time = 5 minutes), the probability density at $x = 3$ minutes.

$$f(3) = 0.2 \cdot e^{-0.2 \times 3} = 0.2 \cdot e^{-0.6} = 0.2 \cdot 0.5488$$
$$= 0.1098.$$

Interpretation

- The relative likelihood of observing a 3-minute interval is ≈ 0.1098.
- Note: For continuous distributions, the PDF value is not a probability but a density. Probabilities are calculated over intervals via integrations.

Exercise with solution

Exercise 1: PDF Evaluation.

Given, $\lambda = 0.1$, mean time = 10 min

$f(5) = ?$, $f(10) = ?$

Soln

(1) $f(5) = 0.1 \times e^{-0.1 \times 5} = 0.1 \cdot e^{-0.5} \approx 0.0607$

(2) $f(10) = 0.1 \times e^{-0.1 \times 10} = 0.1 \cdot e^{-1} \approx 0.0368$.

Observation

→ The density at the mean time is lower than at $x = 0.5$, reflecting the distribution's decreasing nature.

Exercise 2: Probability over an Interval.

For $\lambda = 0.2$ find $P(2 \le x \le 5)$ (probability of waiting between 2 and 5 min)

Soln

$$P(2 \le x \le 5) = F(5) - F(2) = (1 - e^{-0.2 \times 5}) - (1 - e^{-0.2 \cdot 2})$$
$$= e^{-0.4} - e^{-1} = 0.6703 - 0.3678 = 0.3024$$

why $1 - e^{\lambda x}$)

What Problem does it solve?

Imagine you're waiting for a bus that comes randomly but on average every 10 minutes. The exponential CDF answers questions like:

-> What is the probability the next bus arrives within 5 minutes?

- What is the probability the next bus arrives within 5 minutes?
- What is the chance I'll wait more than 15 minutes?

The formula:

$$F(x) = P(\text{Waiting time} \leq x) = 1 - e^{-\lambda x}$$

where,

- λ = Average arrival rate (e.g. 0.1 buses/minute for a 10-minute average wait).
- x = Your chosen time threshold (e.g. 5 minutes).

Bus Waiting Times

Given

- Buses arrive on average every 10 minutes -> $\lambda = \frac{1}{10} = 0.1$ per minute.

Question 1: What's the prob. the next bus arrives within 5 minutes.

Calculation:

$$F(5) = 1 - e^{-0.1 \times 5} = 1 - e^{-0.5} = 1 - 0.6065 =$$

$$= 0.3935 \ (\sim 39\%).$$

Question 2: What's the probability of waiting more than 15 minutes?

$$P(x > 15) = 1 - F(15) = e^{-0.1 \times 15} = e^{-1.5}$$

$$= 0.223 \ (\sim 22.3\%)$$

Why the formula works.

1. $e^{-\lambda x}$: Represent the probability of no events occurring in time x.
   - For our bus: $e^{-0.1 \times 5} = 0.6065$ means there's a 60.65% chance no bus comes in 5 minutes.
2. $1 - e^{-\lambda x}$: Flip this to get the probability of at least one event (bus arrival) in time x.

Exercise 3: Comparing Rates

Compare the PDF shapes for $\lambda = 0.1$ and $\lambda = 0.3$ by:

1. calculate $f(5)$ for both rates.
2. Describing how λ affects the distribution.

Solution:

1.)
- For $\lambda = 0.1$: $f(5) = 0.1 \cdot e^{-0.5} \approx 0.0607$.
- For $\lambda = 0.3$: $f(5) = 0.3 \cdot e^{-1.5} \approx 0.0669$.

2).

Higher λ (e.g. 0.3): shifts the PDF peak toward zero (shorter typical waiting times)

Lower λ (e.g. 0.1): Result in a flatter, wider PDF (longer typical waiting time).

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3. Cumulative Distribution Function (CDF)

The cumulative Distribution Function (CDF) of the exponential distribution describes the probability that the time between events X is less than or equal to a given time x.

1. Mathematical Formulation.

For a rate parameter $\lambda > 0$, the CDF is:

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, \text{ for } x \geq 0.$$

where

- λ = event rate (events per unit time).
- x = time interval.

2. Derivation from the PDF

- The CDF is derived by integrating the PDF $f(x) = \lambda e^{-\lambda x}$

$$F(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}.$$

3. Key properties

(a) Start at 0 : $F(0) = 0$ (impossible to have a waiting time ≤ 0)

(b) Approaches 1 as $x \to \infty$: $\lim_{x \to \infty} F(x) = 1$

Means:

The CDF $F(x) = 1 - e^{-\lambda x}$ approaches 1 as x becomes very large because.

- The term $e^{-\lambda x}$ gets closer to 0 as x increase
- Therefore, $1 - e^{-\lambda x}$ approaches $1 - 0 = 1$

Example

For $\lambda = 0.1$ (mean wait = 10 minutes).

- At $x = 10$ min : $F(10) = 1 - e^{-1} = 0.632$
- At $x = 100$ min : $F(100) = 1 - e^{-10} \approx 0.9999$

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(c) Memoryless Property:

→ Future wait time is independent of past wait time
→ Unique to exponential distribution among continuous distributions.
⇒ formula: $P(X > s+t \mid X > s) = P(X > t)$.

→ if no calls arrived in 10 minutes, the probability of waiting another 5 minutes is the same as waiting 5 min initially.

Example Calculation

Given $\lambda = 0.2$ (mean time between events = 5 minutes).

Compute $F(3)$:

$$F(3) = 1 - e^{-2.5 \times 3} = 1 - e^{-0.6} = 1 - 0.5488$$
$$= 0.4512.$$

Interpretation: There's a 45.12% chance the next event occurs within 3 minutes.

Exercise with Solutions

Exercise 1: Basic CDF Calculation.

For $\lambda = 0.1$ (mean = 10 minutes), Find:

(1) $F(5) = 1 - e^{-0.1 \times 5} = 0.3935$.

(2) $P(X > 15) = 1 - F(15) = e^{-0.1 \times 15} = 0.2231$.

↳ prob of exceeding a wait time a.
↳ connection with Memoryless property.
↳ 1 is cancel out $F(15) = (1 - e^{0.1 \times 15})$

Exercise 2: Comparing Rates.

For two systems.

- System A: $\lambda = 0.3$ (mean = 3.33 minutes).
- System B: $\lambda = 0.1$ (mean = 10 minutes).

Compute $F(5)$ for both and interpret the results.

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Soln

- System A: $F(5) = 1 - e^{-0.3 \times 5} = 0.7769$.
- System B: $F(5) = 1 - e^{-0.1 \times 5} = 0.3935$.

Interpretation: System A has a higher probability. (77.69%) of an event occurring within 5 minutes compared to System B (39.35%), reflecting its faster event rate.

(4. Key Applications)

~~1. Service Call Intervals~~

1. Service Call Intervals.
   - ↳ $\lambda = 0.2$ calls/minute → Average 5 minutes between calls.

2. Equipment Failure
   - ↳ If a machine fails on average every 100 hours ($\lambda = 0.01$), the probability it fails within 50 hours is $1 - e^{\wedge}(-0.01 * 50)$ $= 0.393$.

3. Web Traffic.
   - ↳ Modeling time between website visits.

(5. Relationship with Poisson)

→ Poisson: Counts events per fixed interval.
→ Exponential: Measures time between events.

~~Examples~~.

→ The PDF gives the probability density of the random variable at a specific point x. It tells us how dense the probability is around a particular value of x.

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← PMF x Overview

The PDF of the exponential distribution is defined as:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

where,

- x: The time until the event occurs (a non-negative value).
- λ: The rate parameter (events per unit time), $\lambda > 0$.
- e: Euler's number (approx. 2.71828).

(1) Why PDF and Not PMF?

(a) Continuous Vs. Discrete:

- Exponential distribution is a continuous distribution, meaning it deals with a continuous range of possible values (e.g. time, distance).
- For continuous data, the probability of any exact value x is technically zero. Instead, we talk about the probability over an interval using the area under the curve of the PDF. → $\int_a^a \lambda e^{-\lambda x} dx = 0$

(b) PMF is for Discrete Data:

- PMF is used for discrete random variable, where the outcomes are countable and distinct (e.g. no. of arrivals, no. of failures).
- In contrast, the exponential distribution can take on any positive real value (e.g. 1.2, 3.5, 0.89), not just integer counts.

(c) Probability as Area

the probability is around a particular value of x.

PMF Overview

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(c) Probability as Area

- Since the exponential distribution is continuous, we cannot directly assign a probability to a single

point (like $P(X=x)$).

- Instead, we use the PDF to define a density function and probabilities are obtain by integrating over a ranges of values.

(4) Example:

- in the Poisson distribution (a discrete distribution), we can say

$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- But in the exponential distribution, the probability of a specific time x occuring is zero. Instead, we calculate the probability of the event occurring within a time interval using:

$$P(a \leq x \leq b) = \int_a^b \lambda e^{-\lambda x} dx.$$

This integral gives the area under the PDF curve, representing the probability.

(2) What is Density in the Context of a PDF?

In the context of continuous probability distribution like the exponential distribution, density refers to the relative likelihood of the random variable taking on a particular value.

However, it is not a probability itself but rather a measure of how concentrated the probability is around a particular point.

(3) Understanding Density Visually

(a) For discrete Distributions:

→ in a discrete distribution (like a Poisson distribution), we can assign a specific probability to each individual outcome.

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- For example, $P(X=3) = 0.15$ is a specific probability.

→ For Continuous Distributions:

(a) • In a continuous distribution (like the exponential distribution), the probability of any exact value $X = x$ is zero.

- Instead, we talk about the density at a particular point. The density tell us how concentrated the probability is at that point, but it is not a probability itself.

(b) Density and Area Under the Curve:

→ The height of the PDF curve at any point x is the density.

→ To get a probability, we need to integrate the PDF over an interval.

→ The area under the curve over a specific interval gives the probability for that interval.

For $\lambda = 0.1$ and $x = 5$.

PDF of the exponential distributions,

$$f(x;\lambda) = \lambda e^{-\lambda x}.$$

$$f(5) = 0.1\, e^{0.1 \times 5} \approx 0.06065.$$

This value 0.06065 is the density at $x = 5$, it indicates how concentrated the probability mass is around $x = 5$.

But it is not the probability of $X = 5$. The actual probability requires integrating over a range, such as $[0, 5]$.

(3) Why Calculate Probability Over an Interval?

In a continuous distribution, the probability of a specific point $X = x$ is technically zero. This is because there are infinite many possible values that X can take within any range.

However, we can calculate the probability over an interval $[a, b]$ by integrating the PDF over that range. This gives us the area under the curve, which represents the probability of the random variable falling within that interval.

(a) Example

Suppose we have an exponential distribution with $\lambda = 2$.

→ We want to find the probability that the event occurs between 1 and 3 units of time.

⇒ Thus, we need to evaluate the integral

$$P(1 \leq x \leq 3) = \int_1^3 e^{-2x} \, dx.$$

calculating this integral:

$$= \left[-e^{2x} \right]_1^3$$

$$= -e^{-2 \cdot 3} + e^{-2 \cdot 1}$$

$$= -e^{6} + e^{-2}$$

Evaluating the exponentials

$$= -0.002478 + 0.1353 \approx 0.1328.$$

(b) Why not at a specific Point?

→ If we want to find the probability of the event occurring exactly at $x = 2$, the probability would be zero.

$$P(X = 2) = \int_2^2 \lambda e^{-\lambda x} \, dx = 0.$$

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This is why we focus on intervals rather than specific points. The area under the curve over a range gives us a non-zero probability.

CDF Overview:

(1)

(a) Deriving the the CDF from the PDF

To derive the CDF, we integrate the PDF from 0 to x:

$$F(x) = \int_0^x \lambda e^{-\lambda t}\, dt$$

Step-by-step Integration:

(i) Integrating

$$\int \lambda e^{-\lambda t}\, dt = -e^{-\lambda t}$$

(ii) Applying limits

$$F(x) = -e^{-\lambda t}\Big|_0^x$$

$$= -e^{-\lambda x} + e^0$$

$$= 1 - e^{-\lambda x}$$

Thus, the CDF of the exponential distribution is:

$$F(x) = 1 - e^{-\lambda x}$$

(b) Understanding the Exponential CDF

→ $F(x)$ gives the probability that the waiting time X is less than or equal to x.

→ As $x \to \infty$, $e^{-\lambda x} \to 0$, so $F(x) \to 1$. This aligns with the concept that the probability of the event occurring at some point in time is 100%.

→ As $x \to 0$, $F(x) \to 0$, meaning the probability of the event occurring immediately is 0.

(2) Why does the CDF of the Exponential Distribution use $1 - e^{-\lambda x}$?

(a) Understanding the CDF: What is it?

→ The CDF at a point x is the probability that the random variable X is less than or equal to x:

$$F(x) = P(X \leq x).$$

→ For the exponential distribution, we are dealing with the waiting time until an event occurs.

→ The exponential distribution is memoryless, meaning the probability of waiting a certain amount of time is independent of how much time has already passed.

(b) Deriving the Exponential CDF from the PDF.

The PDF of the exponential distribution is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

To obtain the CDF, we integrate the PDF from 0 to x:

$$F(x) = \int_0^x \lambda e^{-\lambda t} \, dt.$$

Step-by-step Integration:

(a) Integrating the PDF:

$$\int \lambda e^{-\lambda t} \, dt = -e^{\lambda \div}$$

(b) Applying the limits 0 to x:

$$f(x) = -e^{-\lambda t}\Big|_0^x$$

$$= -e^{-\lambda x} + e^{0}$$

$$= 1 - e^{-\lambda x}$$

Thus, the CDF is:

$$F(x) = 1 - e^{-\lambda x}.$$

(c) Why $1 - e^{-\lambda x}$?

→ The exponential distribution models the time until an events occurs, but the CDF represents the probability that the event has already occurred by time x.

→ $e^{-\lambda x}$ represents the probability of no occurance by time x, meaning that event has not yet happened

→ Therefore:

↳ $1 - e^{-\lambda x}$: Probability that the event has occurred by time x.

↳ $e^{-\lambda x}$: Probability that the event has not occurred by time x.

(d) Intuitive Explanation:

→ Suppose we are waiting for a bus that arrives every 10 minutes ($\lambda = 0.1$)

→ The probability that the bus has not arrived by 5 minutes is:

$$P(X > 5) = e^{-0.1 \times 5} = e^{-0.5}$$

$$\simeq 0.6065.$$

→ The probability that the bus has arrived by 5 minutes is

$$1 - e^{0.5} = 1 - 0.6065 = 0.3935$$

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→ Thus, the CDF structure $1 - e^{-\lambda x}$ effective flips the perspective from "no occurance" to "occurence" making it more intuitive for calculating cumulative probabilities.

or

→ can be expressed as

$1 - (\text{Probability of No Event Occuring by time } x)$

Exponential: PDF vs CDF.

① Difference

Aspect	PDF (Probability Density Function)	CDF
Def'n	The PDF $f(x)$ gives the density of occurrences at a specific point x. It is not a probability but measures how concentrated the probability is around x.	The CDF ($F(x)$) gives the accumulated probability up to a point x. It is the probability that the random variable is less than or equal to x.
Formula	$f(x) = \lambda e^{-\lambda x}$ for Exponential	$F(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$
Range	$\lambda \geq 0$	$0 \leq F(x) \leq 1$.
Probability calculation	Requires integrating the PDF over an interval $[a, b]$	Can directly use the CDF at endpoints a and b.

(b) Why They are different but give the same Interval Probability?

~~(a) why~~

The probability of the event occuring within a time interval [a, b] can be expressed in two ways

Pdf Interval

$$P(a \le X \le b) = \int_a^b \lambda e^{-\lambda x}\, dx. \quad \text{---} (1)$$

cdf Interval

$$\text{cdf}, F(x) = 1 - e^{-\lambda x}$$

The probability over the interval [a, b] can be calculated using the CDF as:

$$P(a \le X \le b) = F(b) - F(a).$$
$$= (1 - e^{-\lambda b}) - (1 - e^{-\lambda a}).$$
$$= e^{-\lambda a} - e^{-\lambda b}. \quad \text{---} (II)$$

Integrating eq (1)

Step 1: antiderivative of $\lambda e^{-\lambda x}$ is:

$$-e^{-\lambda x}$$

Step 2: Applying Limits

$$= -e^{-\lambda x} \Big|_a^b.$$
$$= -e^{-\lambda b} + e^{-\lambda a}.$$
$$= e^{-\lambda a} - e^{-\lambda b}. \quad \text{---} (III)$$

Therefore,

$$\int_a^b \lambda e^{-\lambda x}\, dx = F(b) - F(a).$$

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Means: Probability of [a, b] interval

pdf Interval [a, b] = cdf Interval [a, b].

Thus, integrating the PDF over an interval is equivalent to taking the difference of the CDF values at the endpoints. The CDF itself is the accumulated integral of the PDF, so the area calculated through integration is the same as the area obtained using CDF subtractions.

Example Analysis:

Problems

→ The avg waiting time for a customer at a bank is 10 minutes ($\lambda = 0.1$)

→ What is the probability that the next customer will arrive between 5 and 8 minutes?

- Using PDF (Integration)

$$\int_5^8 0.1 e^{0.1x} dx$$

$$= -e^{0.1x} \Big|_5^8 = -e^{0.8} + e^{0.5}$$

$$= 0.4493 - 0.3679 \approx 0.0814.$$

- Using CDF

$$P(5 \le x \le 8) = F(8) - F(5)$$

$$= (1 - e^{-0.8}) - (1 - e^{-0.5})$$

$$= e^{-0.5} - e^{-0.8}$$

$$= 0.4493 - 0.3679$$

$$= 0.0814$$

Memoryless Property.

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(1) What is the Memoryless Property?

→ The memoryless property states that for a given waiting time x is an exponential distribution, the probability of waiting an additional time 't' given that we have already waited for time 's' is independent of 's'.

Mathematically

$$P(X > s + t \mid X > s) = P(X > t).$$

→ The probability of waiting an additional t units of time is the same, regardless of how long you have already waited.

→ In other words, the distribution forgets how much time has already passed.

2) Why is the Exponential Distribution Memoryless?

→ The exponential distribution models time until the first occurrence of an event, like the arrival of a customer or the failure of a machine.

→ Each interval of time is independent of previous intervals.

→ ~~The exponential and geometric distribution are two.~~ The exponential distribution is the only continuous distributions with the memoryless property.

→ The geometric distribution is the discrete counterpart that also exhibits memorylessness.

Exponential (1)

8) Why is it Memoryless?

→ The exponential distribution is memoryless because of its constant hazard rate.

→ The hazard rate λ is the rate at which events occur per unit time.

→ In the exponential distribution, the hazard rate is constant, meaning the likelihood of an event occurring in the next t units of time is always the same, regardless of past waiting time.

→ Mathematically, the exponential CDF is:

$$F(x) = 1 - e^{-\lambda x}$$

→ The survived function (probability of no event occuring up to time x) is:

$$S(x) = e^{-\lambda x}$$

Now, the probability of waiting an additional time t, given that we have already waited s. is:

$$P(X > s+t \mid X > s) = \frac{S(s+t)}{S(s)}$$

$$= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}$$

→ Notice that the term s cancels out, meaning the remaining waiting time is independent of the elapsed time.

4. Intuitive Explanation.

-> Imagine you are waiting for a bus that arrives on average every 10 minutes. You have already waited for 5 minutes, but the bus has not arrived yet.

↳ In a memoryless process, the probability of the bus arriving in the next 5 minutes is the same as it was when you first started waiting - regardless of the fact that you have already waited for 5 minutes.

↳ The clock resets every time. The process does not remember how long you have already waited.

4. Real-life Example:

Example: Machine failure.

-> A machine breaks down on average once every 15 minutes ($\lambda = 1/15$).

-> You have already waited 10 minutes without a breakdown.

-> The probability of a breakdown in the next 5 minutes is:

$$P(X \leq 5) = 1 - e^{(1/15) \times 5} \approx 1 - e^{0.3333}$$

$$= 0.2835$$

-> This probability is the same as if you had just started waiting, despite the 10 minutes already passed. The past is irrelevant.

Does Exponential pdf have memory less property?

→ No, the PDF itself does not have the memoryless property. The memoryless property is a characteristic of the distribution as a whole, not just the PDF. It is specifically a property of the exponential distribution and it is expressed in terms of probabilities or the CDF.

→ Memoryless property involves conditional probabilities, which are derived using the CDF, not the PDF.

→ The PDF $f(x)$ is a function of x, giving the density at a single point. It does not account for elapsed time or remaining time in the context of waiting or occurance. Therefore, it cannot express the concept of "remaining time being independent of elapsed time" which is essential for memory-less.

Survival Function Overview

(1) What is a survival function?

→ ~~is~~ (also known as the reliability function) is a function that gives the probability that a random variable X exceeds a certain value x. Mathematically,

$$S(x) = P(X > x).$$

(2) How is the survival function derived from

→ ~~of~~ the CDF?

→ cdf gives the probability that a random variable X is less than or equal to a specific value x

$$F(x) = P(X \leq x)$$

→ Since the total probability is 1, the probability of the complement (i.e. the event not occuring by time x) is:

$$S(x) = 1 - F(x).$$

→ pdf

$$F(x) = 1 - e^{-\lambda x}.$$

$$S(x) = 1 - (1 - e^{-\lambda x})$$

$$= e^{-\lambda x}$$

→ S(x) gives the probability that the waiting time is greater than x.

→ In other words, it tells us the probability of surviving beyond time x without the event occuring.

(3) no. avant occurring by time x Vs. surviving beyond time x

→ The exponential distribution is memoryless, meaning that the probability of surviving beyond time x is the same as the probability of no avant occuring by time x.

→ Therefore, $e^{-\lambda x}$ both purpose.

- It is the probability of no avant occuring by time x.
- It is also the probability of surviving beyond time x without the avant occurring.

→ There is no contradiction because the survival function $S(x)$ specifically represents the probability of not experiencing the avant by time x., which is mathematically identical to the probability of surviving beyond time x.

→ Both interpretations are essentially the same concept in the context of the exponential distribution.

(4) Examples

→ Imagine a machine that fails on average every 10 minutes ($\lambda = 0.1$).

→ The survival functions $S(x) = e^{0.1 \cdot x}$ tells us that probability that the machine survives without failure beyond x minutes.

Q. What is the probability that the machine survives beyond 15 minutes?

$$S(15) = e^{-0.1 \times 15} = e^{-1.5} \approx 0.2231.$$

→ Thus there is approximately a 22.31% probability that the machine will survive beyond 15 minutes.

PDF

A) Python code PDF

```
import numpy as np.
import matplotlib.pyplot as plt.
- - - - - - Manually - - - - - -
# Exponential PDF Function.
1) def exponential_pdf (x, lambda).
        return lambd * np. exp (-lamb * x)
# Parameters
lambi = 0.1   # Rate parameter
x_value = 5   # Time at which we calculate the PDF

# Calculate PDF at a specific time.
pdf_at_x = exponential_pdf (x_value, lambd)
print (f"PDF at x = {x_value}: {pdf_at_x:.4f}")

# visualization.
x = np. linspace (0, 20, 1000).
pdf_values = exponential_pdf (x, lambda).

plt. figure (figsize = (10,6)).
plt. plot (x, pdf_values, label = 'Exponential
        PDF', color = 'blue').
# Mark the specific point.
plt. scatter (x_value, pdf_at_x, color = 'red',
        zorder = 5)
plt. text (x_value, pdf_at_x, f' ({x_value},
        {pdf_at_x:.4f})', color = 'red').

plt. xlabel ('x')
plt. ylabel ('Density')
plt. title ('Exponential PDF at Specific Time')
plt. legend()
```

plt.legend().
plt.grid(alpha=0.3).
plt.show().

(1) def exponential_pdf(x, lambd):
    return lambd * np.exp(-lambd * x)

--- Using builtin function ---
from scipy.stats import expon

lambda = 0.1
x_value = 5.

Calculate PDF using SciPy.
pdf_at_x = expon.pdf(x_value, scale=1/lambda)
print(f"PDF at x = {x_value}: {pdf_at_x:.4f}")

Note:
- The scale parameter is 1/λ
- In our case, 1/0.1 = 10.

Manually uses the rate parameter (λ), not the scale parameter.

→ Manually takes the rate parameter lambd directly and calculates the PDF using the formula:

$$f(x) = \lambda e^{-\lambda x}$$

→ This is the standard rate-based form of the exponential PDF

→ The custom function does not need to convert λ to β because it directly uses the scale parameter in its formula. If we wanted to write a function that behaves like scipy and accepts the scale parameter instead, we would need to modify it as follow.

```
def exponential_pdf_scale(x, scale):
    return (1/scale) * np.exp(-x/scale)
```

(3) Python Code: Exponential pdf Interval Probability.

```
import numpy as np.
import matplotlib.pyplot as plt.
from scipy.integrate import quad.

# Exponential PDF function.
def exponential_pdf(x, lambd):
    return lambd * np.exp(-lambd * x).

# Parameters.
lambd = 0.1  # Rate parameter (1/mean)
interval = (5, 10) # Interval over which we
                   calculate the probability.
# Calculate probability over the interval
  using pdf.
probability, _ = quad(exponential_pdf,
    interval[0], interval[1], args=
    (lambd,)).
```

```
print(f"Probability over interval {interval}
      : {probability:.4f}").

# Visualization:
x = np.linspace(0, 20, 1000).
pdf_values = exponential_pdf(x, lambd)

plt.figure(figsize=(10,6)).
plt.plot(x, pdf_values, label='Exponential
         PDF', color='blue').

# Highlight the interval area.
x_fill = np.linspace(interval[0], interval
                     [1], 1000)
pdf_fill = exponential_pdf(x_fill, lambd)
plt.fill_between(x_fill, pdf_fill, color
    ='orange', alpha=0.5, label
    =f'Interval {interval}')

plt.xlabel('x')
plt.ylabel('Density')
plt.legend()
plt.grid(alpha=0.3).
plt.show()
```

(1) Why exponential_pdf is not explicitly called before passing it to quad()?

→ quad() is a function from scipy.integrate used to perform numerical integration

Syntax:

quad(func, a, b, args=())

→ Understanding functions:

↳ in Python, function are first-class objects, meaning they can be:

→ Passed as argument to other function
→ Assigned to variable
→ Returned from other functions.

→ How does quad() call exponential_pdf()?

↳ quad() will call exponential_pdf() internally, repeatedly, over small subintervals to calculate the integral.

↳ It will pass each value of 'x' to exponential_pdf() along with the additional arguments (lambd).

↳ It will pass each value of x to exponential_pdf() along with the additional arguments (lambd,)

→ Why not save the Return Value?

↳ The purpose of passing the function without calling it is to allow quad() to evaluate it at multiple points within the interval.

↳ If you were to call exponential-pdf()
directly passing it to quad(), it would
only return a single value, not a function
that can be evaluated across an interval.

~~quad(exponential-pdf(a, λ), a, b~~

quad(exponential-pdf, interval[a], interval
[b], args = (lambd,))

Vs

quad(exponential-pdf(x, λ), " "
, "))

→ a single call vs. functions calls.

(1) why a single call fails?

→ exponential-pdf(5, 0.1) is evaluated immediately
→ This returns a single float value not a function
→ we end up passing something like:

result = quad(exponential-pdf(5, 0.1), 5, 10,
args=(0.1,))

result = quad(0.060653, 5, 10, args=(0.1,))

→ Here, 0.060653 is not callable. Thus, quad()
will throw a TypeError, expecting a
function a function but receiving a float.

(2) Why Passing the function works?

→ When we pass the function reference, we are giving quad() the ability to call the function multiple times over the interval

result = quad(exponential_pdf, 5, inf, args = (0, 1))

→ Here, exponential_pdf is not called immediately.

→ Instead, it is passed as a function reference to quad()

Now, quad() can call it as many times as necessary to approximate the area under the curve.

Analogy

→ Think of quad() as a painter who need to paint a fence made up of multiple sections.

→ If you pass a single float value, it's like giving the painter one specific brush stroke -- it's useless for painting the entire fence.

→ But when you pass the function reference, it's like giving the painter the entire brush, allowing them to paint every section of the fence.

(2) x_fill = np.linspace(interval[0], interval[1], 1000)

pdf_fill = exponential_pdf(x_fill, lambd).

plt.fill_between(x_fill, pdf_fill, color='orange', alpha=0.5, label=f'{interval}')

→ creates 1000 points between 5 and 10.
↳ These x values will form the ox-axis of the shaded area.

→ x_fill (Numpy array), the function is applied element-wise, calculating the PDF values for each x in the intervals.
↳ This gives us the y-axis values for the shaded area.

→ plt.between() fills the area between a curve and x-axis.

Finding pdf Interval using Builtin

```
# Parameters
lambd = 0.1
mean = 1/lambd
interval = (5, 10).

#using the inbuilt PDF function.
x = np.linspace (0, 20, 1000)
pdf_values = expon.pdf (x, scale=mean)
        (or) #(scale = 1/λ = mean).

#Calculate probability over the interval
using CDF
probability = expon.cdf (interval[1],
        scale=mean) - expon.cdf
        (interval[0], scale=mean).
print (f" Probability over interval
        {interval}: {probability:
        .4f}")).
```

Break down

probability Calculations

↳ The probability over the interval [a, b] is calculated using the CDF method:

$$P(a \leq x \leq b) = P(b) - F(a).$$

(2) Python Program to calculate Exponential CDF.

```
# Manually
import numpy as np.

def exponential_cdf(x, lam):
    """
    Calculate the CDF of the exponential
    distribution at a given x

    Parameters:
    - x (float) : Time at which to calculate
                  the CDF.
    - lam (float) : Rate parameter (1/mean)

    Returns:
    - float : CDF value at x
    """
    if x < 0:
        return 0 # CDF is 0 for negative x
    return 1 - np.exp(-lam * x).

# Example Parameters
x = 5
lam = 0.1

# Calculate CDF.
cdf_value = exponential_cdf(x, lam).

print(f"The CDF value at x = {x} with
      λ = {lam} is: {cdf_value:.4f}")
```

output

The CDF value at x = 5 with λ = 0.1 is:
0.3935.

Built-in Function

```
from scipy.stats import expon
scale = 1/lam

cdf_value = expon.cdf(x, scale=scale)
```

Break-down

(1)
```
def exponential_cdf(x, lam):
    if x < 0:
        return 0
    return 1 - np.exp(-lam * x)
```

→ The exponential distribution is defined only for non-negative values of x.

→ The probability of an event occurring before time 0 is logically impossible. so the CDF is 0 for negative.

→ The CDF of the exponential distribution is given by

$$F(x) = 1 - e^{-\lambda x}$$

(3) Python code: Memory less Property of Exponential Distribution: Understanding through a Real-world Example.

```
import numpy as np

def exponential_survival(t, lam):
    """Survival function"""
    return np.exp(-lam * t)

def memoryless_property(s, t, lam):
    """
    Demonstrate memoryless property:
    P(X > s+t | X > s) == P(X > t)
    """
    # P(X > s+t)
    prob_st = exponential_survival(s+t, lam)

    # P(X > s)
    prob_s = exponential_survival(s, lam)

    # Conditional probability P(X > s+t | X > s)
    cond_prob = prob_st / prob_s

    # Independent probability P(X > t)
    prob_t = exponential_survival(t, lam)

    return cond_prob, prob_t

# Parameters
lam = 0.5
s = 5  # already waited time in minutes
t = 10  # Additional waiting time in minutes
```

```
# Calculate probabilities
cond_prob, prob_t = memoryless_property
                    (s, t, lam)

# Display results.
print(f"Probability of waiting at
    least {t} more minutes after
    already waiting {s} minutes:
    {cond_prob:.4f}").
print(f"Probability of waiting of.
    waiting at least {t} minutes
    from the start: {prob_t:.4f}")

# Verifying Memoryless Property
print("\nMemoryless Property Verified:
    "if np.isclose(cond_prob, prob_t)
    else "Memoryless Property Not
    Verified").
```

output:

→ Probability of waiting at least 10 more minutes after already waiting 5 minutes : 0.3679.

→ Probability of waiting at least 10 minutes from the start: 0.3679.

→ Memoryless Property Verified:

Python Code: Survival function.

```
Import numpy as np
From scipy.stats import expon.

#Manual.
def manual_sf (x, lam).
    """
    s(x) = e^(-λx).
    """
    return np.exp (- lam * x)

#Parameters.
lam = 0.1 # Rate parameter (1/mean), meaning
          1 event every 10 minutes
time_to_check = 15 # Time at which we calculate
                   the survival probability.

# Calculate survival probability using both
  methods.
sf_manual = manual_sf (time_to_check, lam)

# Built in.
sf_scipy = expon.sf (time_to_check,
                     scale = 1/lam).

# Display result.
print (f"Manual survival function at
       {time_to_check} mins: {sf_manual
       :.4f}").
print (f"Scipy survival function at
       {time_to_check} mins: {sf_scipy:
       .4f}").
```

output

0.2231

Generalized Linear Model (GLM)

~~2:00~~

Real-world Analogy:

Scenario: Coffee shop Sales.

You're managing a coffee shop and want to predict the number of orders (cups of coffee sold per hour)

You believe the number of coffee orders depends on:

1. Number of visitors
2. Whether it's a rainy day (people order more coffee when it rains).
3. Time of day (e.g. more sales in the morning)

So you collected this data hourly

- Visitors = 20
- Is Raining = 1 (yes)
- Hour = 9 AM
- Orders = 18

You want to predict orders using these inputs. That's what GLM does.

- Input Features = Visitors, Is_Raining, Hour
- Output (Target) = Orders (count data)

pure Math Explanation of GLM

1. Linear Predictor

We start by forming a linear equation.

$$\eta = \beta_0 + \beta_1 \cdot \text{Visitors} + \beta_2 \cdot \text{IsRaining} + \beta_3 \cdot \text{Hour}$$

This part is just like a linear regression. Let's assume.

- $B_0 = 0.5$
- $B_1 = 0.1$
- $B_2 = 0.6$
- $B_3 = 0.05$

For:

Visitors = 20, Is Rainy = 1, Hour = 9.

$\eta = 0.5 + 0.1 \cdot 20 + 0.6 \cdot 1 + 0.05 \cdot 9 = 3.55$

what is Beta (β)?

Beta (β) values are the coefficients (weights) that the GLM adjusts during training to make the predicted value ($\hat{y}$) as close as possible to the actual observed value (y).

(2) link functions

We use a link function to convert this linear output into a valid prediction for count data.

For Poisson GLM, the link function is exponential

$$\mu = \exp(\eta)$$

So,

$$\mu = \exp(3.55) \approx 34.82$$

This means: Based on the conditions, we expect 34.82 coffee orders.

(B) Poisson Distribution Assumption

we assume the actual observed orders come from a Poisson distribution with mean μ.

$$y \sim \text{Poisson}(\mu = 34.82).$$

so, if we observe only 18 orders, the model uses this mismatch (error) to adjust the βs in training using maximum likelihood estimation (MLE).

Summary:

GLM is like using a dial-based machine to tune the weight (β) on your predictions (visitor, rain, time) until the predicted orders match actual orders, assuming your counts follow a Poisson law.

Mathematically

The Poisson GLM models the mean number of events (λ) as a log-linear function of predictor variables.

$$y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki}$$

- y_i: observed count (e.g., number of orders in hour i)
- λ_i: expected count (Poisson mean)
- $x_{1i}, x_{2i}, \dots$ predictor variables (e.g., visitors, hour, flash sale indicator)

$\log(\cdot)$: link function (ensures $\lambda > 0$).

This form applies here,

$$\log(x) = y$$

$$\boxed{x = e^y}$$

python code

```
from statsmodels.formula.api import glm
from statsmodels.api import families

model = glm(
    formula = 'orders ~ visitors + is_flash
                - scale - hour + C(hour)',
    data = df_hrly,
    family = families.Poisson()
).fit()
```

dhawalsingh

Condition	PMF/CDF Method (Poisson-based)	Z-score Method (counts vs. mean/std)	GLM Poisson			GLM Negative Binomial
			Z-score	Absolute Threshold (counts > pred + x)	Relative Threshold (counts > 1.5x pred)	
$\lambda < 5$ (highly Discrete, Non-Normal)	☑ Valid but may be too sensitive (low PMF values common)	☒ Not valid (skewed, discrete)	⚠ misleading std-resid	☑ works but threshold tuning needed.	☑ Better than z-score	☑ Best - handles dispersion and skew
$\lambda \geq 10$ (Near-Normal Smooth)	☑ Valid	☑ Valid	☑ Valid	☑ Valid	☑ Valid	☑ Best - generalizes better.
No overdispersion Mean ≈ Variance	☑ Valid	☑ Valid	☑ Valid	☑ Valid	☑ Valid	☑ works equally well
Over dispersion Variance > Mean	~~☑ Valid~~ ✗ Invalid (assumes fixed λ)	~~☑ Valid~~ ✗ Misleading (Underestimate variance)	~~☑~~ ✗ Standardized residuals biased	⚠ Threshold-based may misfire	⚠ Arbitary scaling	☑ Correctly account for extra variance

Here is a comprehensive summary table evaluating each spike detection methods across different data conditions.

Kolmogorov-Smirnov (KS) Test

→ The Kolmogorov-Smirnov (KS) test is a statistical test that compares a sample distribution to a theoritical distribution (like exponential, normal etc) to check how well they match.

What does it do?

It test the hypothesis:

" is the data likely to have come from this distribution.

For your churn model:

- ~~Your churn model~~
- You are saying: " I believe customer lifetimes follow an Exponential distribution
- KS tells you: How close your data is to ~~be~~ an exponential distribution.

How it works (Conceptually)

(a) Problem setup.

You have:

→ A set of n observed data points.

$x_1, x_2, x_3 \ldots x_n$

You also have:

- A theoritical cumulative distribution function (CDF): $F(x)$.

(b) Empirical CDF (ECDF)

This is the CDF built from your data (no formulas, just counts).
Defined as:

$$F_n(x) = \frac{\text{Number of points} \leq x}{n}$$

For example, if 3 out of 10 points are ≤ 5 then $F_n(5) = 0.3$.

(c) KS Test statistics (D)

The core idea of the KS test is to find the maximum distance between the observed and expected CDFs:

$$D_n = \sup_x |F_n(x) - F(x)|$$

Where:

- $F_n(x)$ = ECDF (your data)
- $F(x)$ = CDF of your model (e.g. exponential).
- $\sup_x$ = maximum over all x
- D_n = KS statistics.

In plain words:
-) Find the largest gap between the CDF of your data and your model's CDF

(d) p-value.

Once you compute D_n, the p-value is calculated using the KS distribution of KS values under the null hypothesis

- if $p > 0.05 \Rightarrow$ model is okay.
- if $p \leq 0.05 \rightarrow$ model is not a good fit.

This is usually computed using libraries (like scipy.stats.kstest) because it involves complex integrals.

Examples

We are given 4 customer lifetimes (in months).

Data: [2, 3, 4, 5]

We want to check if these come from an Exponential Distribution with $\lambda = 1/3$ (i.e. mean = 3).

Soln

(1) Sort the data

$X = [2, 3, 4, 5]$

No. of samples (n) = 4

(2) Empirical CDF:

Empirical CDF at each point is:

$$F_n(x_i) = \frac{i}{n}$$

x_i	$\hat{F}_n(x_i)$
2	$1/4 = 0.25$
3	$2/4 = 0.50$
4	$3/4 = 0.75$
5	$4/4 = 1.00$

3. Theoratical CDF of Exponential ($\hat{\lambda} = 1/3$):

$$F(x) = 1 - e^{-\hat{\lambda}x}$$

Compute for each x_i:

x_i	$F(x_i)$	Value
2	$1 - e^{2/3}$	$= 0.4866$
3	$1 - e^{1}$	$= 0.6321$
4	$1 - e^{4/3}$	$= 0.7364$
5	$1 - e^{5/3}$	$= 0.8262$.

4. Compute difference

$$D_i = |F_n(x_i) - F(x_i)|$$

x_i	$F_n(x_i)$	$F(x_i)$	D_i
2	0.25	0.4866.	0.2366.
3	0.50	0.6321	0.1321.
4	0.75	0.7364	0.0136.
5	1.00	0.8262	0.1738.

Final KS statistics

$$D = \max(D_i) = \max(0.2366, 0.1321, 0.0136, 0.1738) = \boxed{0.2366}$$

You compare this D-value with a value from a KS table, or calculate a p-value if:

- D > critical value, we reject the null hypothesis (doesn't follow exponential)
- D ≤ Critical value, we fail to reject the null (data fits Exponential well).

Python Code: Manual & Built-in ks Test

```
import numpy as np
from scipy.stats import expon, kstest.

# Sample data.
lifetimes = np.array([1, 2, 3, 4])
lambda_hat = 1/np.mean(lifetimes)

# Step1: sort data
sorted_lifetimes = np.sort(lifetimes).

# Step2: Empirical CDF
empirical_cdf = np.arrange(1, len(sorted_
                lifetimes)+1)/len(sorted_
                lifetimes)
# arrange(start, stop) -> stop at n+1 = 4+1 = 5
# Step3: Theoritical CDF using fitted exponential
  distribution.
theoritical_cdf = expon.cdf(sorted_lifetimes,
                  scale = 1/lambda_hat).
```

#Step4: Differances

differances = amirical - theoretical_cdf
abs_diff = np.abs(differance)

#Step 5: Manual ks statistics.
ks_stat_manual = np.max(abs_diff)

#Step6: Built in ks test.
ks_stat_builtin, p_value = kstest(lifetimes, 'expon', arg = (0, 1/lambda_hat)).

How D-critical are determined?

D-critical is the threshold value for the KS test statistic (D) at a certain significance level (like 0.05). It depends.
-) The sample size 'n'
-) The significance level α (commonly 0.05 to 0.01)
-) Whether the test in one-sample or two-sample

Formula,

$$D_{critical} = \frac{C(\alpha)}{\sqrt{n}}$$

where:
- n = sample size
- $c(\alpha)$ = constant depending on significance level (e.g., for $\alpha = 0.05$, $c \approx 1.36$)

Significance Level (α)	$c(\alpha)$ Constant.
0.10	1.22
0.05	1.36
0.01	1.63.

so if $n = 60$ and $\alpha = 0.05$, then

$$D_{critical} = \frac{1.36}{\sqrt{60}} \approx 0.192$$

~~2nd~~

~~P-value Interpretation.~~

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